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From Intelligent Tools to Autonomous Agents: A Conceptual Framework for Agentic AI in Organizational Systems

Muhammad Ajmal

Department of Management Science, University of Gujrat, Gujrat, Pakistan.

Email: ajmal.hailian@gmail.com

Mehreen Ashraf

Department of Business Administration, Hajvery University Lahore.

drmehreen.ashraf@hup.edu.pk

Azmat Islam*

Department of Business Administration, University of Education, Lahore, Pakistan.

Corresponding Author Email: azmat24@gmail.com

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ABSTRACT

The evolution of artificial intelligence (AI) from task-specific tools to autonomous, goal-directed agents marks a fundamental shift in how organizations design, manage, and execute work. This article develops a conceptual framework for understanding *agentic AI* in organizational systems, defined as AI systems capable of independent decision-making, adaptive learning, and coordinated action across complex environments. Moving beyond traditional automation and augmentation paradigms, the framework positions agentic AI as an active organizational actor that reshapes workflows, decision rights, and inter-functional coordination. The paper systematically examines the integration of agentic AI across core organizational functions, including operations, marketing, finance, and human resources, highlighting its role in enabling real-time optimization, continuous learning, and decentralized decision-making. It further identifies key dimensions of agentic AI adoption, including autonomy, adaptability, goal alignment, and human-AI collaboration. Building on these dimensions, the study proposes a multi-layered framework that links technological capabilities with organizational structures, governance mechanisms, and strategic outcomes. In addition, the article addresses critical challenges associated with agentic AI, such as transparency, accountability, ethical alignment, and control. It argues that the successful deployment of agentic AI requires a reconfiguration of organizational practices, including the development of new governance models, capability-building initiatives, and trust-building mechanisms. By synthesizing insights from AI, management, and information systems literature, this conceptual study contributes to theory-building by offering an integrative perspective on how autonomous AI agents can be embedded within organizational systems. The framework provides a foundation for future empirical research and offers practical guidance for organizations seeking to leverage agentic AI for sustainable competitive advantage.

Keywords: Agentic Artificial Intelligence; Autonomous Agents; Organizational Systems; Conceptual Framework; Human-AI Collaboration; Organizational Governance; Intelligent Automation; Digital Transformation; Decision-Making; AI Strategy

1. Introduction

Artificial intelligence (AI) has undergone a profound transformation over the past decade, evolving from rule-based systems and

predictive analytics into more advanced forms characterized by autonomy, adaptability, and goal-directed behavior. This shift has given rise to *agentic artificial intelligence (agentic AI)*—a class of systems capable of independently perceiving environments, making decisions, and executing actions to achieve specified or emergent objectives (Islam, Ajmal, & Islam, 2025) . Unlike traditional AI tools that primarily support human decision-making, agentic AI systems function as semi-autonomous or autonomous actors within organizational contexts, fundamentally altering how work is performed and managed (Russell & Norvig, 2021; Wooldridge, 2020).

The emergence of agentic AI is particularly significant in the context of modern organizations, which operate in increasingly dynamic, data-intensive, and uncertain environments (Ajmal, Islam, & Khalid, 2025). The growing complexity of global markets, supply chains, and customer ecosystems demands faster, more adaptive, and continuously optimized decision-making processes. Traditional organizational models—often characterized by hierarchical decision structures and functionally siloed operations—are increasingly inadequate for such conditions (Ajmal, Khan, & Islam, 2024). Agentic AI offers a pathway to address these challenges by enabling decentralized, real-time decision-making and continuous process optimization across organizational functions (Raisch & Krakowski, 2021).

Across functional domains, the applications of agentic AI are rapidly expanding. In operations and supply chain management, intelligent agents can autonomously monitor disruptions, reconfigure logistics networks, and optimize inventory flows in real time (Ivanov & Dolgui, 2021). In marketing, AI agents dynamically personalize customer interactions, adapt campaigns based on real-time behavioral data, and enhance customer engagement through continuous learning (Huang & Rust, 2021). In finance, agentic systems are employed in algorithmic trading, fraud detection, and risk assessment, often operating with minimal human intervention (Baker & Dellaert, 2018). Similarly, in human resource management, AI-driven agents support talent acquisition, performance management, and workforce analytics, enabling more data-driven and adaptive people management practices (Jarrahi, 2018).

While these developments highlight the transformative potential of agentic AI, they also signal a deeper shift in the nature of organizational systems (Ajmal, Khalid, & Islam, 2025). Organizations are no longer composed solely of human actors but

increasingly include intelligent, autonomous agents that participate in decision-making and execution processes (Ajmal, 2022). This hybrid configuration challenges traditional assumptions about organizational design, authority, and coordination. For instance, the allocation of decision rights between humans and AI agents raises questions about accountability, control, and trust. Moreover, the integration of agentic AI necessitates new forms of collaboration, where humans and machines jointly contribute to problem-solving and innovation (Raisch & Krakowski, 2021; Jarrahi, 2018).

Despite its growing relevance, the concept of agentic AI remains underdeveloped in the organizational and management literature (Ajmal, Islam, & Islam, 2025). Existing research has largely focused on AI as a tool for automation or augmentation, emphasizing efficiency gains and decision support. However, agentic AI introduces a qualitatively different paradigm in which AI systems possess a degree of autonomy and agency, requiring a rethinking of organizational structures, processes, and governance mechanisms. There is a need for integrative frameworks that conceptualize how agentic AI can be embedded across organizational functions and how it reshapes interdependencies among them (Islam, Khalid, & Ajmal, 2025).

In addition to theoretical gaps, the adoption of agentic AI presents significant practical and ethical challenges. As AI systems gain autonomy, issues related to transparency and explainability become more complex, making it difficult for organizations to fully understand and justify AI-driven decisions (Doshi-Velez & Kim, 2017). Concerns about bias, fairness, and ethical alignment are amplified when systems operate with limited human oversight (Floridi et al., 2018). Furthermore, the potential risks associated with unintended behaviors or misaligned objectives highlight the importance of robust governance frameworks and control mechanisms. Organizations must therefore balance the benefits of autonomy with the need for accountability and ethical responsibility (Ajmal, Islam, & Islam, 2024).

Another critical dimension of agentic AI adoption is its impact on the workforce and organizational capabilities. Rather than replacing human workers entirely, agentic AI is more likely to transform roles, shifting the focus from routine task execution to oversight, strategic thinking, and collaboration with intelligent agents. This transformation requires the development of new skills, including AI literacy, data interpretation, and the ability to manage

human-AI interactions effectively (Jarrahi, 2018). It also necessitates cultural and structural changes within organizations to foster trust, adaptability, and innovation (Malik, Bakri, Ajmal, & Malik, 2019).

Given these complexities, there is a pressing need for a comprehensive conceptual framework that captures the multifaceted nature of agentic AI in organizational systems. Such a framework should integrate technological, organizational, and managerial perspectives, providing insights into how agentic AI capabilities interact with organizational structures, processes, and governance mechanisms. It should also address the cross-functional implications of agentic AI, recognizing that its impact extends beyond individual departments to the organization as a whole.

This article seeks to address this gap by developing a conceptual framework for agentic AI in organizational systems. Specifically, it aims to (1) define and conceptualize agentic AI within the organizational context, (2) examine its applications and implications across key organizational functions, (3) identify the core dimensions and capabilities that characterize agentic AI systems, and (4) propose governance and design principles for their effective integration. By synthesizing insights from the fields of artificial intelligence, management, and information systems, this study contributes to theory development and provides a foundation for future empirical research.

In doing so, the article advances the understanding of how organizations can transition from using AI as a passive tool to leveraging it as an active, agentic participant in organizational systems. It argues that the successful integration of agentic AI requires not only technological innovation but also a reconfiguration of organizational logic, encompassing structure, strategy, and governance. Ultimately, this work aims to inform both scholars and practitioners on how to harness the transformative potential of agentic AI while addressing its inherent challenges.

2. Literature Review

The emergence of *agentic artificial intelligence (AI)* builds upon several streams of research spanning artificial intelligence, information systems, and organizational theory. This section synthesizes prior literature across four key domains: (1) evolution of AI in organizations, (2) concept of agency and intelligent agents, (3) AI across organizational functions, and (4) governance, ethics, and human-AI collaboration. The review highlights the theoretical

foundations and identifies gaps that motivate the development of a conceptual framework for agentic AI in organizational systems.

2.1. Evolution of AI in Organizational Contexts

Early applications of AI in organizations were largely rule-based systems designed to automate structured, repetitive tasks. These systems relied on predefined logic and lacked the ability to learn or adapt beyond their programming (Russell & Norvig, 2021). With the advent of machine learning and big data analytics, AI systems evolved into predictive tools capable of identifying patterns and supporting decision-making processes. This shift enabled organizations to enhance efficiency and accuracy across functions such as forecasting, customer segmentation, and risk analysis (Brynjolfsson & McAfee, 2017).

More recent developments emphasize AI as a means of *augmentation* rather than mere automation. In this paradigm, AI systems complement human capabilities by providing insights, recommendations, and decision support. Raisch and Krakowski (2021) describe this as the “automation-augmentation paradox,” where AI simultaneously replaces and enhances human labor. Similarly, Jarrahi (2018) highlights the concept of human-AI symbiosis, where decision-making is distributed between humans and intelligent systems.

However, the latest wave of AI innovation extends beyond augmentation toward *autonomy*. Agentic AI systems are characterized by their ability to act independently, set sub-goals, and adapt to changing environments. This transition marks a significant departure from earlier models, positioning AI as an active participant in organizational processes rather than a passive tool.

2.2. Theoretical Foundations of Agency and Intelligent Agents

The concept of agency in AI is rooted in the field of multi-agent systems, where an *agent* is defined as an entity that perceives its environment and acts upon it to achieve specific goals (Wooldridge, 2020). Russell and Norvig (2021) further conceptualize intelligent agents as systems that maximize expected utility through rational decision-making. These definitions provide the foundational basis for understanding agentic AI.

In organizational theory, agency is traditionally associated with *agency theory*, which examines relationships between principals (e.g., managers) and agents (e.g., employees) (Jensen & Meckling, 1976). The introduction of AI agents complicates this framework by introducing non-human actors capable of

autonomous decision-making. This raises questions about control, monitoring, and incentive alignment, as AI agents do not possess intentions in the human sense but operate based on programmed objectives and learned behaviors.

Recent studies suggest that agentic AI requires a reconceptualization of agency within organizations. Instead of viewing agency solely as a human attribute, organizations must consider hybrid systems where human and artificial agents jointly contribute to outcomes. This perspective aligns with socio-technical systems theory, which emphasizes the interdependence of social and technical components in organizational design (Bostrom & Heinen, 1977).

2.3. AI Across Organizational Functions

AI technologies have been widely adopted across various organizational functions, each with distinct applications and implications.

In **operations and supply chain management**, AI enables real-time monitoring, predictive maintenance, and adaptive logistics. Ivanov and Dolgui (2021) demonstrate how digital supply chain twins can simulate disruptions and support autonomous decision-making, enhancing resilience and efficiency.

In **marketing**, AI facilitates customer analytics, personalization, and engagement. Huang and Rust (2021) propose a strategic framework in which AI progresses from mechanical tasks to analytical and ultimately intuitive functions, enabling deeper customer understanding and dynamic interaction.

In **finance**, AI applications include algorithmic trading, credit scoring, and fraud detection. These systems often operate with high levels of autonomy, making rapid decisions based on complex data inputs (Baker & Dellaert, 2018). The increasing reliance on AI in financial decision-making underscores the importance of accuracy, transparency, and risk management.

In **human resource management**, AI supports recruitment, performance evaluation, and workforce planning. Jarrahi (2018) notes that AI systems can enhance decision-making by providing data-driven insights, but they also require human oversight to ensure fairness and contextual understanding.

While these applications demonstrate the functional benefits of AI, most existing research focuses on isolated use cases rather than cross-functional integration. Agentic AI, by contrast, operates across boundaries, coordinating activities and decisions in a more holistic manner.

2.4. Human-AI Collaboration and Organizational Implications

The integration of AI into organizational processes has significant implications for human roles and interactions. Rather than replacing humans entirely, AI often reshapes work by redistributing tasks between humans and machines. Jarrahi (2018) emphasizes the importance of *human-AI symbiosis*, where humans provide contextual judgment and ethical reasoning, while AI contributes computational power and data processing capabilities.

Raisch and Krakowski (2021) further argue that organizations must balance automation and augmentation to achieve optimal outcomes. Excessive reliance on automation may reduce human involvement and oversight, while insufficient use of AI may limit efficiency gains. Agentic AI intensifies this tension by introducing systems that operate with greater independence, requiring new forms of coordination and trust.

Trust is a critical factor in human-AI collaboration. Users must have confidence in the reliability and fairness of AI systems to effectively integrate them into decision-making processes. However, as AI systems become more complex and opaquer, achieving transparency and explainability becomes increasingly challenging (Doshi-Velez & Kim, 2017).

2.5. Governance, Ethics, and Control of Agentic AI

The rise of agentic AI raises important questions about governance and ethical responsibility. Traditional governance mechanisms, which rely on human oversight and hierarchical control, may be insufficient for managing autonomous systems. Organizations must develop new frameworks that ensure accountability, transparency, and alignment with organizational goals.

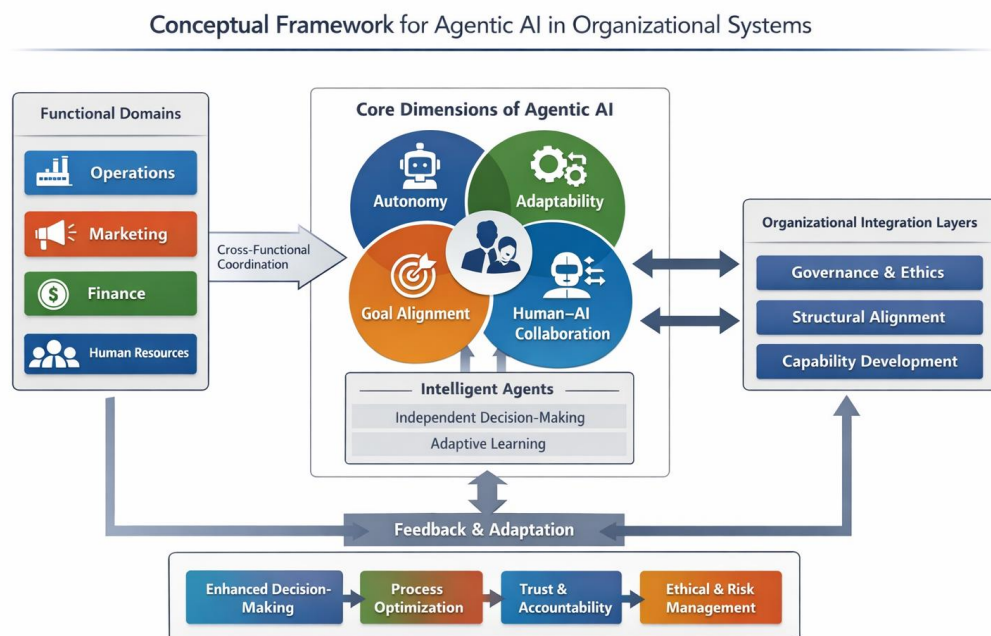
Floridi et al. (2018) propose an ethical framework for AI that emphasizes principles such as beneficence, non-maleficence, autonomy, justice, and explicability. These principles are particularly relevant for agentic AI, where decisions may have significant and far-reaching consequences. Ensuring that AI systems operate in accordance with these principles requires careful design, monitoring, and regulation.

Another key challenge is *goal alignment*, which refers to ensuring that AI systems pursue objectives consistent with organizational and societal values. Misalignment can lead to unintended behaviors or outcomes, highlighting the importance of robust control mechanisms and continuous oversight (Russell & Norvig, 2021).

2.6. Research Gaps and Theoretical Contribution

Despite the growing body of literature on AI in organizations, several gaps remain. First, existing research largely focuses on AI as a tool for automation or augmentation, with limited attention to its role as an autonomous agent. Second, studies often examine AI applications within specific functions, neglecting the cross-functional integration that characterizes agentic AI. Third, there is a lack of comprehensive frameworks that integrate technological capabilities with organizational structures, governance mechanisms, and strategic outcomes.

This article addresses these gaps by developing a conceptual framework for agentic AI in organizational systems. It integrates insights from AI, management, and information systems literature to provide a holistic understanding of how agentic AI can be effectively embedded across organizational functions. In doing so, it contributes to theory-building and lays the foundation for future empirical research.



3. Conceptual Model Explanation: Agentic AI in Organizational Systems

3.1. Functional Domains as the Foundation of Agentic AI Integration

The model begins with key organizational functions—operations, marketing, finance, and human resources—which serve as the primary contexts for the deployment of agentic artificial

intelligence (AI). These domains represent traditionally siloed areas of organizational activity that are increasingly interconnected through digital transformation. The integration of AI agents across these functions enables real-time data exchange, enhanced coordination, and improved decision-making processes. Research indicates that cross-functional AI implementation significantly improves organizational performance, leading to higher efficiency, better forecasting accuracy, and increased return on investment compared to isolated AI applications (Aziz et al., 2025). Thus, functional domains act as the entry points through which agentic AI initiates organizational transformation.

3.2. Core Dimensions of Agentic AI

At the center of the model are four defining dimensions of agentic AI: autonomy, adaptability, goal alignment, and human-AI collaboration. Autonomy refers to the ability of AI systems to make independent decisions and perform tasks without continuous human intervention, distinguishing agentic AI from traditional rule-based systems (Islam et al., 2025). Adaptability reflects the system's capacity to learn from data and dynamically adjust its behavior in response to environmental changes, thereby enhancing resilience and responsiveness (Malatji, 2025). Goal alignment ensures that AI systems operate in accordance with organizational objectives, minimizing risks associated with misaligned actions and improving strategic coherence (Bandara et al., 2026). Human-AI collaboration emphasizes the importance of hybrid systems in which human judgment complements AI-driven insights, ensuring contextual awareness and ethical oversight (Zárate-Torres et al., 2025). Collectively, these dimensions form the conceptual backbone of agentic AI.

3.3. Intelligent Agents as the Execution Mechanism

The model conceptualizes intelligent agents as the operational layer through which agentic AI capabilities are enacted. These agents perform tasks, make decisions, and continuously optimize processes across operational, tactical, and strategic levels. Unlike traditional automation systems, intelligent agents exhibit learning and reasoning capabilities, enabling them to handle complex and dynamic tasks. Empirical findings suggest that such agents enhance decision quality, reduce operational costs, and improve efficiency by leveraging large-scale data processing and predictive analytics (Shahzaib et al., 2025). This execution layer represents the transformation of AI from a passive tool into an active organizational participant.

3.4. Organizational Integration Layers

The successful implementation of agentic AI requires alignment with organizational structures, which is captured in the model through three integration layers: governance and ethics, structural alignment, and capability development. Governance and ethics are essential for ensuring transparency, accountability, and responsible AI usage, particularly given the risks associated with autonomous decision-making systems (Sumbaraju, 2025). Structural alignment involves embedding AI systems into organizational processes and decision hierarchies, often leading to more decentralized and flexible organizational structures (Hughes et al., 2025). Capability development focuses on equipping employees with the necessary skills to interact effectively with AI systems, including data literacy and AI management competencies (Khan et al., 2025). These layers ensure that agentic AI is not only technologically feasible but also organizationally sustainable (Ajmal, Manzoor Dar, Islam, & Islam, 2025).

3.5. Feedback and Adaptation Mechanism

A key feature of the model is the feedback and adaptation loop, which reflects the dynamic and evolving nature of agentic AI systems. Through continuous feedback, AI agents learn from outcomes and refine their decision-making processes over time. This aligns with the conceptualization of autonomous systems as adaptive socio-technical entities that evolve through interaction with their environment (Beck et al., 2022). The feedback mechanism also enables organizations to adjust governance frameworks, operational strategies, and system configurations based on AI-driven insights, thereby fostering continuous improvement and organizational learning (Islam, Ajmal, & Khalid, 2025).

3.6. Strategic Outcomes of Agentic AI Implementation

The model culminates in a set of strategic outcomes, including enhanced decision-making, process optimization, trust and accountability, and ethical and risk management. Agentic AI enables faster and more accurate decision-making, improves operational efficiency, and enhances organizational agility (Shahzaib et al., 2025; Islam et al., 2025). At the same time, it necessitates the development of robust governance frameworks to ensure transparency and maintain stakeholder trust (Raza et al., 2025). Ethical and risk management considerations are particularly critical, as autonomous systems can introduce biases, privacy concerns, and accountability challenges if not properly managed (Odedina, 2023). These outcomes highlight both the opportunities

and challenges associated with agentic AI adoption.

3.7.Theoretical Contribution of the Model

Overall, the model presents agentic AI as a central coordinating intelligence that transforms organizations into adaptive, data-driven, and hybrid human-AI systems(Ajmal, Islam, & Khalid, 2025). It extends existing literature by positioning AI not merely as a tool but as an autonomous organizational actor. Furthermore, it integrates technological capabilities with organizational structures and strategic outcomes, offering a comprehensive framework for understanding the role of agentic AI in modern organizations. The model underscores that successful implementation depends not only on technological advancements but also on organizational readiness, governance mechanisms, and continuous learning processes.

4. Discussion

The purpose of this study was to develop a comprehensive conceptual framework for understanding the role of agentic artificial intelligence (AI) in organizational systems. The proposed model extends existing literature by positioning AI not merely as a technological tool but as an autonomous, adaptive, and goal-directed organizational actor(Khalid, Islam, & Ajmal, 2025). This discussion elaborates on the theoretical contributions, managerial implications, and emerging challenges associated with agentic AI, while also outlining directions for future research.

4.1.Theoretical Implications

This study makes several important contributions to the literature on AI and organizational theory. First, it advances the conceptualization of AI by shifting from the dominant perspectives of automation and augmentation toward *agency*. While prior research has emphasized AI as a support system for human decision-making, agentic AI introduces a paradigm in which systems possess autonomy, proactiveness, and decision-making capabilities (Islam et al., 2025). This aligns with emerging research that frames AI agents as active participants capable of reshaping organizational structures and processes (Hughes et al., 2025).

Second, the model contributes to the understanding of organizations as **hybrid socio-technical systems**. Traditional organizational theory has largely focused on human actors; however, the integration of agentic AI necessitates a broader perspective that includes artificial agents as part of the organizational fabric. Autonomous systems increasingly take on roles that involve decision rights and task execution, thereby

redistributing authority within organizations (Beck et al., 2022). This reconceptualization challenges existing theories of coordination, control, and hierarchy, suggesting the need for new frameworks that account for human-AI co-agency.

Third, the study integrates multiple dimensions—technological, organizational, and strategic—into a unified framework. By linking core AI capabilities (autonomy, adaptability, goal alignment, and collaboration) with organizational integration mechanisms and outcomes, the model provides a holistic view of agentic AI. This integrative approach responds to calls in the literature for more comprehensive frameworks that bridge fragmented research across AI applications and organizational contexts (Benmamoun et al., 2025).

4.2. Managerial Implications

From a managerial perspective, the findings highlight that the successful implementation of agentic AI requires more than technological adoption; it demands **organizational transformation**. One key implication is the need for **cross-functional integration**. Organizations that deploy AI across multiple functions—such as operations, marketing, finance, and human resources—are more likely to achieve synergistic benefits, including improved efficiency and higher returns on investment (Aziz et al., 2025). This suggests that managers should move beyond siloed AI initiatives and adopt enterprise-wide strategies.

Another important implication concerns **governance and ethical oversight**. As AI systems become more autonomous, the risks associated with decision-making—such as bias, lack of transparency, and accountability gaps—become more pronounced. Effective governance frameworks are therefore essential to ensure responsible AI deployment (Sumbaraju, 2025). Organizations must establish clear guidelines, monitoring mechanisms, and ethical standards to maintain trust and compliance.

The model also underscores the importance of **human-AI collaboration**. Rather than replacing human workers, agentic AI transforms roles by shifting responsibilities toward supervision, strategic decision-making, and collaboration with intelligent systems. Leadership plays a critical role in facilitating this interaction by balancing algorithmic efficiency with human judgment and ethical considerations (Zárate-Torres et al., 2025). Consequently, organizations must invest in **capability development**, including AI literacy, data skills, and interdisciplinary collaboration.

Finally, the presence of a **feedback and adaptation loop** suggests that organizations should adopt a continuous learning approach. Agentic AI systems improve over time through data-driven feedback, and organizations must be prepared to adapt their processes, structures, and strategies accordingly. This dynamic capability is essential for maintaining competitiveness in rapidly changing environments.

4.3. Practical Challenges and Risks

Despite its potential benefits, the adoption of agentic AI presents several challenges. One major concern is **transparency and explainability**. As AI systems become more complex and autonomous, understanding how decisions are made becomes increasingly difficult, which can undermine trust and accountability (Raza et al., 2025). Organizations must therefore invest in explainable AI techniques and transparent decision-making processes.

Another challenge relates to **ethical and regulatory issues**. Autonomous AI systems can introduce biases, privacy risks, and unintended consequences if not properly governed. Ensuring ethical alignment requires not only technical solutions but also organizational policies and cultural change (Odedina, 2023). Additionally, regulatory frameworks are still evolving, creating uncertainty for organizations adopting advanced AI technologies.

Workforce transformation is also a critical issue. While agentic AI can enhance productivity, it may also lead to job displacement and skill gaps. Research suggests that organizations must proactively address these challenges through reskilling and upskilling initiatives to prepare employees for new roles in AI-enabled environments (Khan et al., 2025).

Finally, **integration complexity** remains a barrier. Many organizations struggle to move beyond pilot projects due to challenges related to data integration, system compatibility, and organizational resistance. Effective implementation requires not only technological infrastructure but also strong leadership and strategic alignment (Bandara et al., 2026).

4.4. Future Research Directions

The proposed framework opens several avenues for future research. First, empirical studies are needed to validate the relationships proposed in the model, particularly the impact of agentic AI dimensions on organizational outcomes. Quantitative research using structural equation modeling or longitudinal designs could provide deeper insights into causal mechanisms.

Second, future research should explore the **role of context**, including industry differences, organizational size, and cultural factors, in shaping agentic AI adoption. Different sectors may exhibit varying levels of readiness and face distinct challenges, which warrants comparative analysis.

Third, there is a need to further investigate **human-AI interaction dynamics**, particularly in terms of trust, decision-making authority, and collaboration. Understanding how humans and AI agents co-create value will be critical for designing effective hybrid systems.

Finally, research should focus on developing **robust governance frameworks** that balance autonomy with control. This includes exploring new models of accountability, ethical decision-making, and regulatory compliance in the context of agentic AI.

5. Conclusion of Discussion

In conclusion, this study highlights that agentic AI represents a fundamental shift in how organizations operate, moving from tool-based automation to autonomous, intelligent systems that actively participate in decision-making and execution. While the benefits of agentic AI are substantial, its successful implementation requires careful consideration of organizational design, governance, and human-AI collaboration. By providing a comprehensive conceptual framework, this research contributes to both theory and practice, offering a foundation for future studies and guiding organizations in navigating the complexities of agentic AI adoption.

References

- Ajmal, M. (2022). Ethical leadership and employee creative performance: Discussing the mediating role of employer feedback environment in software houses of Pakistan. *Reviews of Management Sciences*, 4(1), 192-210.
- Ajmal, M., Islam, A., & Islam, Z. (2024). Unraveling the path to organizational commitment: The mediating role of perceived organizational support and the moderating role of career stages in high-performance work systems. *Global Business Review*, 25(6).
- Ajmal, M., Islam, A., & Khalid, S. (2025). A Socio-Technical Systems Perspective on Organizational Performance: Integrating Soft Systems Methodology and High-Performance Work Systems. *ASSAJ*, 3(02), 2672-2688.
- Ajmal, M., Islam, A., & Khalid, S. (2025). The future of organizations: Moving from knowledge management toward organizational consciousness through knowledge transcendence. *Research*

- Consortium Archive*, 3(3), 1735-1738.
- Ajmal, M., Islam, Z., & Islam, A. (2025). Enhancing organizational performance in higher education through knowledge-centered culture and absorptive capacity: The mediating role of the knowledge creation process. *The Learning Organization*, 32(5), 733-756.
- Ajmal, M., Khalid, S., & Islam, A. (2025). A systems-based perspective on organizations: Integrating soft systems methodology and knowledge management systems. *Journal of Business and Management Research*, 4(5), 177-191.
- Ajmal, M., Khalid, S., & Islam, A. (2025). Organizational problem solving as a conscious process: Integrating soft systems methodology and organizational consciousness. *Journal of Business and Management Research*, 4(4).
- Ajmal, M., Khan, I., & Islam, A. (2024). Fostering employee creativity in software houses: Exploring the influence of ethical leadership, LMX, and employee feedback-seeking behavior. *Migration Letters*, 21(S9), 1-18.
- Ajmal, M., Manzoor Dar, W., Islam, A., & Islam, Z. (2025). Empowering Collective Intentionality: A Framework for Social Change Through Group Consciousness. *World Futures*, 81(1), 15-34.
- Aziz, F., Muzaffar, F., Shahid, S., Ahmed, H. S., & Iqbal, S. M. (2025). The role of artificial intelligence in driving ROI through synergized HR, marketing, and financial decision-making. *Inverge Journal of Social Sciences*.
- Baker, T., & Dellaert, B. (2018). Regulating robo-advice across the financial services industry. *Iowa Law Review*, 103(2), 713-750.
- Bandara, E., Gore, R., Shetty, S., Rajapakse, S., Kularathna, I., Karunarathna, P., et al. (2026). A practical guide to agentic AI transition in organizations.
- Beck, R., Dibbern, J., & Wiener, M. (2022). A multi-perspective framework for research on (sustainable) autonomous systems. *Business & Information Systems Engineering*, 64, 265-273.
- Benmamoun, Z., Benhamida, H., & Jizat, J. M. (2025). The integration of AI and business excellence in infrastructure and development: A literature review and directions for future research. *Journal of Infrastructure, Policy and Development*.
- Bostrom, R. P., & Heinen, J. S. (1977). MIS problems and failures: A socio-technical perspective. *MIS Quarterly*, 1(3), 17-32.
- Brynjolfsson, E., & McAfee, A. (2017). *Machine, Platform, Crowd*:

- Harnessing Our Digital Future*. W.W. Norton & Company.
- Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*.
- Floridi, L., Cowls, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., ... Vayena, E. (2018). AI4People—An ethical framework for a good AI society. *Minds and Machines*, 28(4), 689–707.
- Huang, M.-H., & Rust, R. T. (2021). A strategic framework for artificial intelligence in marketing. *Journal of the Academy of Marketing Science*, 49(1), 30–50.
- Hughes, L., Dwivedi, Y. K., Malik, T., Shawosh, M., Albashrawi, M., Jeon, I., et al. (2025). AI agents and agentic systems: A multi-expert analysis. *Journal of Computer Information Systems*.
- Islam, A., Ajmal, M., & Islam, Z. (2025). Unlocking work engagement: Harnessing social capital, proactive personalities and POS in newcomers' organizational socialization. *International Journal of Organizational Analysis*, 33(8), 1916–1942.
- Islam, A., Ajmal, M., & Khalid, S. (2025). Beyond knowledge management: Reframing organizations as knowledge ecologies for a wisdom-based management paradigm. *Pakistan Journal of Social Science Review*, 4(7), 786–798.
- Islam, A., Khalid, S., & Ajmal, M. (2025). From complexity to clarity: Merging soft systems thinking with knowledge transcendence in modern organizations. *Journal of Business and Management Research*, 4(3).
- Islam, M. A., Somu, S., & Aldaihani, F. (2025). The rise of agentic AI: Synthesis of current knowledge and future research agenda. *Global Business and Organizational Excellence*.
- Ivanov, D., & Dolgui, A. (2021). A digital supply chain twin for managing disruptions and resilience. *Production Planning & Control*, 32(9), 775–788.
- Jarrahi, M. H. (2018). Artificial intelligence and the future of work: Human-AI symbiosis in organizational decision making. *Business Horizons*, 61(4), 577–586.
- Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4), 305–360.
- Khalid, S., Islam, A., & Ajmal, M. (2025). From academic freedom to algorithmic agency: Knowledge governance in AI-enhanced education. *Journal of Management Science Research*

- Review*, 4(3), 1036-1058.
- Khan, S. R., Baig, M. A., Qaiser, A., Abrar, F., & Ali, H. (2025). Humans and machines at work: Redefining strategic human resource management in the era of artificial intelligence and workforce automation. *The Critical Review of Social Sciences Studies*.
- Malatji, M. (2025). A cybersecurity AI agent selection and decision support framework. *arXiv*.
- Malik, S., Bakri, N., Ajmal, M., & Malik, B. (2019). Role of work motivation as moderator between breach of psychological contract and workplace deviant behavior among employees. *International Journal of Academic Research in Business and Social Sciences*, 9(9), 1036-1046.
- Odedina, E. A. (2023). Redefining governance, risk, and compliance (GRC) in the digital age: Integrating AI-driven risk management frameworks. *World Journal of Advanced Engineering Technology and Sciences*.
- Raisch, S., & Krakowski, S. (2021). Artificial intelligence and management: The automation-augmentation paradox. *Academy of Management Review*, 46(1), 192-210.
- Raza, S., Sapkota, R., Karkee, M., & Emmanouilidis, C. (2025). TRiSM for agentic AI: A review of trust, risk, and security management in LLM-based agentic multi-agent systems. *arXiv*.
- Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
- Shahzaib, R., Latif, F., & Ali, S. W. (2025). The role of intelligent AI agents in optimizing business management and decision-making processes. *Research Journal for Social Affairs*.
- Sumbaraju, A. C. (2025). Agentic AI meets data governance: The unseen battle for control in customer data platforms. *International Journal of Computing and Engineering*.
- Wooldridge, M. (2020). *An Introduction to MultiAgent Systems* (2nd ed.). Wiley.
- Zárate-Torres, R., Rey-Sarmiento, F., Acosta-Prado, J., Gómez-Cruz, N. A., Yaneth, D., Castro, R., & Camargo, J. (2025). Influence of leadership on human-artificial intelligence collaboration. *Behavioral Sciences*, 15.